

Aplikasi MCSA dalam implementasi Predictive Maintenance

Basic Theory for Electrical Motor

PENGERTIAN

MESIN LISTRIK :

- Mengubah tenaga mekanik ke tenaga listrik = Generator
- Mengubah tenaga listrik ke tenaga mekanik = Motor Listrik
- Mengubah tenaga listrik ke tenaga listrik = Transformator

**SEMUA PERISTIWA INI MENGGUNAKAN
MEDAN MAGNET SEBAGAI MEDIA**

Ciri-ciri Tiap Tipe Motor Listrik

Deskripsi Motor Listrik :

Mesin listrik yang mengubah tenaga listrik menjadi tenaga mekanik

Terbagi atas :

- DC Motor (Arus searah)
- AC Motor (Arus bolak-balik)
 - INDUKSI (ASINKRON)
 - SINKRON

CIRI CIRI :

Motor arus searah:

- Kutub magnet di stator
- Arus d.c masuk ke rotor
- Menggunakan slip ring
- Menggunakan exiter

Motor sinkron:

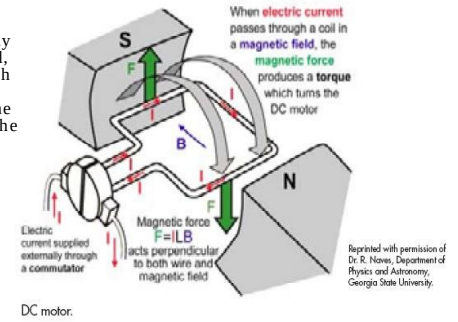
- Kutub magnet di rotor
- Dapat digunakan sebagai kapasitor

Motor induksi:

- Tidak ada magnet nyata

Ciri-ciri Tiap Tipe Motor Listrik

- In the direct-current (DC) motor, the magnetic field is usually stationary and the field, set-up by armature with current-carrying conductors, rotates. The current is supplied to the armature through a commutator and brushes.

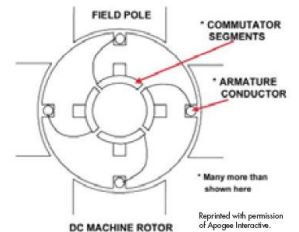


DC motor.

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Ciri-ciri Tiap Tipe Motor Listrik

- **Field pole** – The purpose of this component is to create a steady magnetic field in the motor. For the case of a small DC motor, a permanent magnet, field magnet, composes the field structure. However, for larger or more complex motors, one or more electromagnets, which receive electricity from an outside power source, is/are the field structure.
- **Armature** – When current goes through the armature, it becomes an electromagnet. The armature, cylindrical in shape, is linked to a drive shaft in order to drive the load. For the case of a small DC motor, the armature rotates in the magnetic field established by the poles, until the north and south poles of the magnets change location with respect to the armature. Once this happens, the current is reversed to switch the south and north poles of the armature.
- **Commutator** – This component is found mainly in DC motors. Its purpose is to over-turn the direction of the electric current in the armature. The commutator also aids in the transmission of current between the armature and the power source.

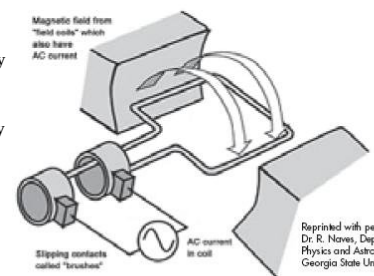


Parts of a direct current motor.

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Ciri-ciri Tiap Tipe Motor Listrik

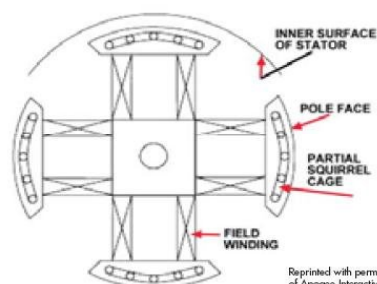
- In the alternating-current induction motor, the currents in the rotor are supplied by induction. The rotor's conductors are cut by the stators' alternating magnetic field (set up by the alternating-current supply to the fields), which induces alternating current in the armature conductors.



AC motor.

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Ciri-ciri Tiap Tipe Motor Listrik



Parts of an alternating current motor.

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Ciri-ciri Tiap Tipe Motor Listrik

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Rotor

- Induction motor

Two types of rotors are used in induction motors:

=> squirrel cage rotor and wound rotor.

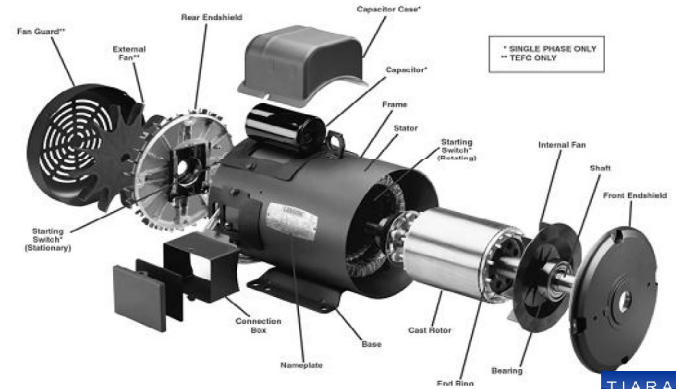
A squirrel-cage rotor consists of thick conducting bars embedded in parallel slots. These bars are short-circuited at both ends by means of short-circuiting rings.

A wound rotor has three-phase, double-layer, distributed winding. It is wound for as many poles as the stator. The three phases are wye internally and the other ends are connected to slip-rings mounted on a shaft with brushes resting on them.

- Synchronous motor – The main difference between the synchronous motor and the induction motor is that the rotor of the synchronous motor travels at the same speed as the rotating magnetic field. This is possible because the magnetic field of the rotor is no longer induced. The rotor either has permanent magnets or DC-excited currents, which are forced to lock into a certain position when confronted with another magnetic field.

Komponen Motor Listrik

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Single Phase AC Motor (Courtesy of Leeson Electric, a Subsidiary of Regal-Beloit Corporation).

Ciri-ciri Tiap Tipe Motor Listrik

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Stator

- Induction motor – The stator is made up of a number of stampings with slots to carry three phase windings. It is wound for a definite number of poles. The windings are geometrically spaced 120 degrees apart.

- Synchronous motor – The stator produces a rotating magnetic field that is proportional to the frequency supplied.

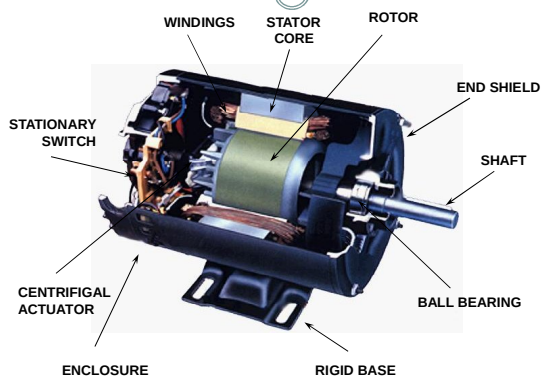
Komponen Motor Listrik

14

- Frame - The frame is a housing that holds all internal and external motor components.
- External Fan - Positioned on the exterior end of the motor assembly, the fan blows air across the outside of the motor to prevent overheating.
- Fan Cover - Protects the fan from foreign object damage and protects people from injury.
- Front Bearing - Shaft support that reduces friction rotation. The front bearing is located at the end where the shaft extends from the motor housing.
- Cast Rotor – Also know of as the armature. The armature is the assembly of windings. These windings interact with the stator to form the magnetic fields necessary for generating rotation energy.
- End Ring – Located at the end of the rotor assembly, the end rings provide a surface to connect the rotor bars.
- Stator – Stationary portion of the motor's electrical field.

Konstruksi Motor Listrik

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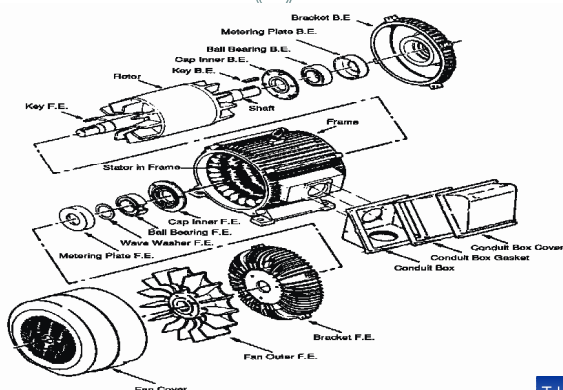
Komponen Motor Listrik

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- Rear Bearing - Shaft support that reduces friction rotation. The rear bearing is located at the same end as the external cooling fan.
- Base – Also referred to as the foot mount, the base secures the motor housing to the equipment platform.
- Shaft – The shaft runs longitudinally through the center of the stator, and is supported by the front and rear bearings.
- Front End Shield – Located at the end where the shaft extends from the motor, the front end shield encloses the motor frame and supports the front bearing.
- Rear End Shield – Located at the same end as the external cooling fan, the rear end shield encloses the motor frame and supports the rear bearing.
- Internal Fan – Located within the motor housing, the internal fan circulates air internally

Komponen Motor Listrik

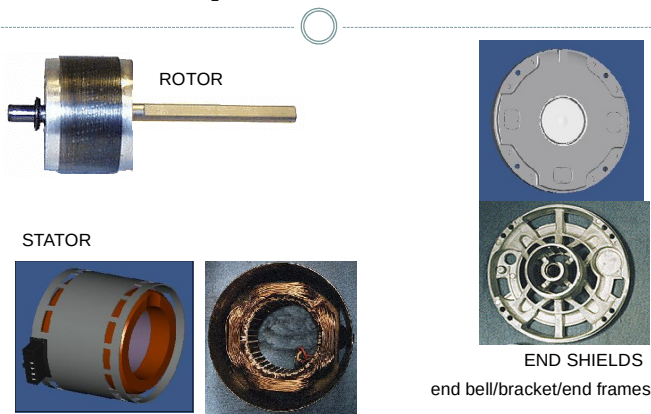
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Komponen Motor Listrik

- Capacitor – Affixed to the outside of the motor, the capacitor shifts the phase of the incoming current and feeds the starter winding.
- Nameplate – A tag affixed to the motor to designate motor specification parameters.
- Connection Box – A box affixed to the motor housing where motor connections are made with the power supply.

Komponen Motor Listrik



BEARINGS

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There are two types of bearings in general use in fractional horsepower motors.

A **ball bearing** is used in motors subject to high thrust or axial loads. Ball bearings cannot be relubricated. The normal lubricant is grease. Farm duty motors use a grease with low-temperature properties because most of the motors are subject to extreme low ambient temperature.

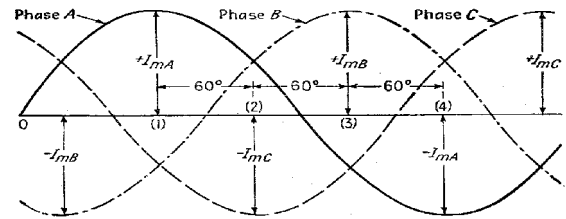
A **sleeve bearing** is used in motors subject to moderate thrust or axial loads. This includes many of the applications we are familiar with. Sleeve bearings are considered less noisy, but with the quantity of today's ball bearings, the noise difference in most applications is minimal. Sleeve bearings can usually be relubricated.

TIARA

The Revolving Field

21

- Three independent currents flow in a three phase induction motor.
- The currents are displaced from each other by exactly 120 electrical degrees.



TIARA

Theory of Operation

22

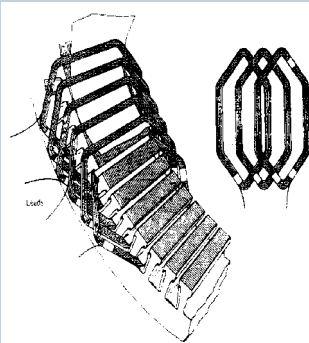
- The conversion of electrical energy into mechanical energy takes place in the rotor. The conversion of power is transferred across the air gap inductively. The relative motion between the revolving field and the rotor causes the rotor bars to be cut by the lines of flux. This generates a rotor voltage causing current to flow in the rotor. The rotor then rotates due to the developed torque. The difference between the revolving field and the rotor speed is called slip.

TIARA

Stator Windings

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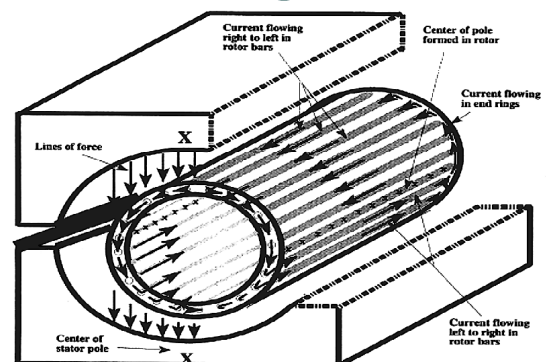
- Comprised of turns of copper wire that form a coil.
- The coils are placed in the slots of the core and separated from one another and the core by insulation.
- The coils are then connected to form a phase.



TIARA

Theory of Operation

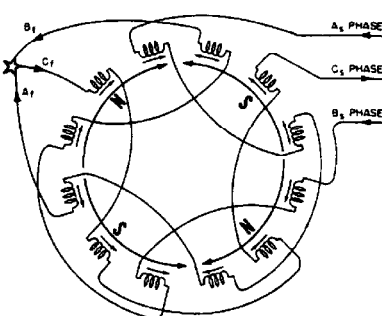
23



TIARA

3 Phase, 4 Pole Winding Diagram

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TIARA

Alternating Current –AC

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- Alternating current => arus bolak-balik => secara interval waktu tetap berubah di satu titik positive dan negative.
- Medan magnet putar dibuat => membuat tenaga 3 phase.
- Stator dihubungkan sumber tenaga listrik 3 phase, masing2 dibuat terpisah pada sirkuit yang sama. Satu putaran = 360 derajat, sehingga masing2 phase dibuat berbeda 120 derajat.
- Gambar dibawah ini menggambarkan kurva sinus.



Listrik 3 phase

Gerak putaran

TIARA

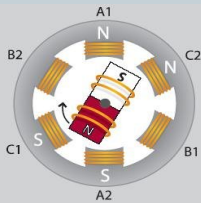
Perubahan kutub

25

Stator dan rotor berinteraksi => motor dapat berputar. Untuk memperlihatkan gerakan secara jelas, rotor kita gantikan dengan magnet batang. Rotor berputar bebas karena pengaruh medan putar dari stator. Gambar motor listrik 2 kutub 3 phase.

Phase dihubungkan berpasangan seperti motor sebenarnya :

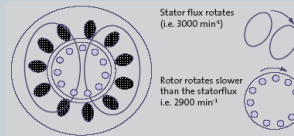
- phase 1 terdiri dari A1 dan A2
- phase 2 terdiri dari B1 dan B2
- phase 3 terdiri dari C1 dan C2



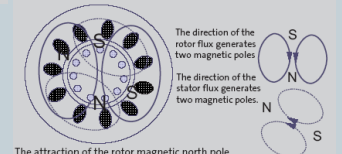
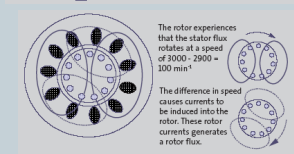
Rotating magnet field dalam Gerakan putaran lambat

Rotating Magnetic Field

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- Rotating stator flux ditimbulkan oleh medan putar magnet stator
- yang dibentuk oleh arus listrik dari tiap phase



The attraction of the rotor magnetic north pole

Atraksi magnet kutub utara rotor terhadap kutub selatan stator dan menghasilkan kekuatan kopel antara stator dan rotor. Kekuatan merupakan torsi motor yang menghasilkan putaran rotor

Aplikasi AC Motor

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- Phase winding / kumparan A,B dan C disusun berbeda 120 derajat. Jumlah kutub ditentukan oleh jumlah winding dipasang perphase, jika winding itu dua maka motor itu berarti motor 2 kutub, demikian juga bila 4 winding itu motor 4 kutub.
- Ketika arus listrik dialirkan ke motor putaran masing2 motor tentu berbeda-beda sesuai dengan jumlah kutub.



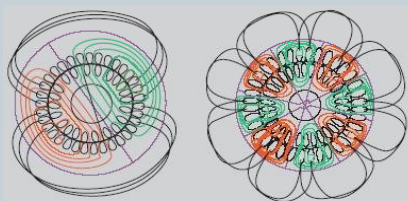
Application of MCSA, flux coil for Electrical Motor Condition Monitoring (Part 1)

- Limitation of application
- Building Database using RBMwizard Program on AMS Suite Software
- Route Management and Analyzer Data Transfer
- Data Collection

Rotating Magnetic Field

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- Medan magnet putar

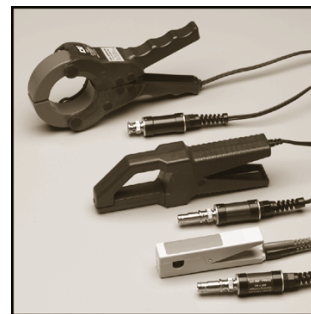


3 phase, 2 kutub motor

3 phase 8 kutub motor

MCSA – Motor Current Signature Analysis

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Volt Adapter

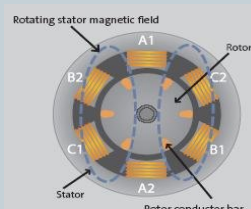


CSI Model 2120-2

Induksi magnet

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- didepan kita menggambarkan rotor dengan menggunakan magnet permanen, yang berputar didalam stator.
- Sebetulnya dalam prakteknya rotor tidak menggunakan magnet biasa. Analog dari stator sama terhadap rotor, yaitu menggunakan medan magnet yang di-induksi dari stator ke konduktor bar rotor, sehingga terjadi polarisasi rotor.



Ketika stator dialiri arus, menimbulkan medan magnet dan memotong konduktor rotor dan menginduksi arus rotor

MCSA – Motor Current Signature Analysis

32



- RB1 - phase 1 current signature
- (Hi-resolution 1600-3200 lines)

- RB2 - phase 2 current signature

- (Low resolution 100 lines)

- RB3 - phase 3 current signature

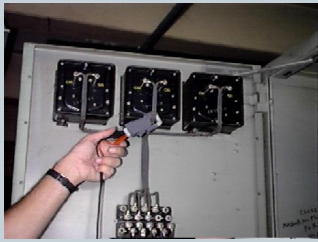
- (Low resolution 100 lines)

Fmax : 80 Hz

Beban motor harus >50%load
Data diambil setiap tahun or semi annual
Terutama motor start/stop, beban fluktuasi

MCSA – Motor Current Signature Analysis

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Limit Aplikasi MCSA :

- Harus mengakses panel listrik
- Ammeter dihubungkan kabel listrik
- Membutuhkan electrician untuk membantu
- Survey:
 - Hanya 10% kerusakan motor disebabkan oleh rotor problems

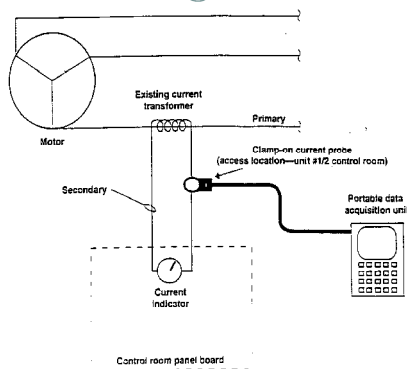
Parameter tingkat kerusakan rotor

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Category Number	dB	A/B	Rotor Condition	Corrective Action
1	>60	>1000	Excellent	None
2	54-60	500-1000	Good	None
3	48-54	250-500	Moderate	Trend
4	42-48	125-250	Rotor Bar May be Cracked or High Resistance Joints	Increase Monitoring Time
5	36-42	63-125	Two Rotor Bars Cracked and High Resistance Joints	Vibration Testing
6	30-36	32-63	Multiple Rotor Bars Broken, Slip Ring and Joint Problems	Overhaul
7	<30	<32	Severe Problems Throughout	Overhaul / Replace

MCSA – Motor Current Signature Analysis

34



Contoh bad rotor

38

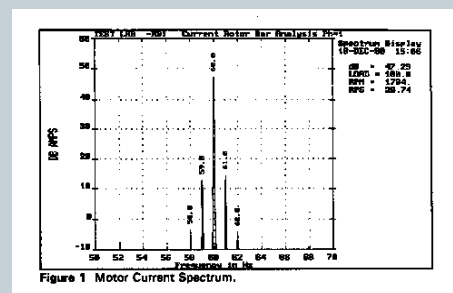


Figure 1 Motor Current Spectrum.

Parameter tingkat kerusakan rotor

35

- Fr = motor rotational frequency
- LSB1 = lower slip sideband of line frequency
- Lf = line frequency
- NP = number of poles
- SF = slip frequency
- $LSB1 = Lf - (NP * SF)$

Perhitungan estimasi jumlah broken rotor

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- Rumus dari Thompson dan Rankin (estimasi konservatif)

$$n = \frac{2R}{10^{\left(\frac{\Delta dB}{20}\right)} + NP}$$

n = estimasi jumlah broken rotor bars
 ΔdB = selisih dB amplitude antara Line Frequency dan LSB 1
 NP = jumlah stator poles
 R = jumlah rotor bars atau slots

Parameter tingkat kerusakan rotor

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- Selisih dB amplitude antara LF dan LSB1
- 54 dB- Kondisi motor normal.
- 45- 54 dB- Kondisi motor adalah marginal condition
- 40- 45 dB- Setidaknya satu broken rotor bar dapat ditemukan (lakukan trending dan monitor terus)
- <40 dB- Lebih dari satu broken rotor bar. Direkomendasikan untuk mematikan motor bila dalam kondisi ini.
- Reference : Electrical Apparatus Service Association (EASA)

Contoh Perhitungan

40

- Dari gambar current spectrum

Lf amplitude = 47.3 dB
 LSB1 amplitude = 14.1 dB
 AIP = 4 stator poles
 R = 58 rotor bars

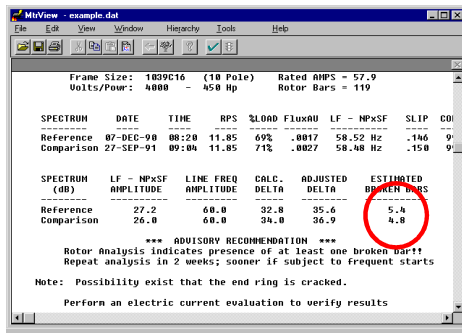
- Jumlah bad rotor bar

$$n = \frac{2 \times 58}{10^{\left(\frac{47.3-14.1}{20}\right)} + 4} = \frac{116}{10^{1.66} + 4} = \frac{116}{45.71} = 2.5 \text{ broken rotor bars}$$

- Hasil pemeriksaan ternyata ditemukan 4 buah broken rotor bar

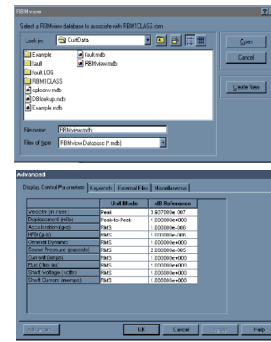
Contoh

41

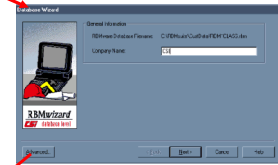


Pembuatan Database Menggunakan RBMwizard

45



- Ketika muncul window RBMview, ketik file name "**motor current**" RBMview.mdb dan klik **Create New**
- Pada Company Name, tulis nama perusahaan



- Klik **Advanced**, Pilih line frekuensi 50 Hz (60 Hz) pada **Miscellaneous**.

Pembuatan Database Menggunakan RBMwizard pada AMS Suite Software

42

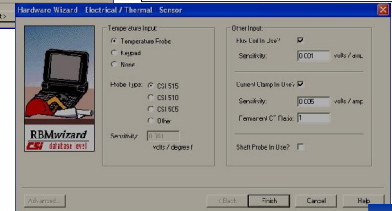
MOTOR CURRENT ANALYSIS

Pembuatan Database Menggunakan RBMwizard

46



- Pada screen technology pilih **electrical/thermal**
- Pada Other input isi **sensitivity flux coil**, dan **Current Clamp** yang digunakan, lalu klik **Finish**



Tujuan :

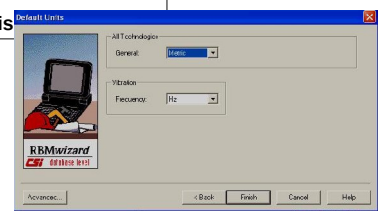
43

- Mempelajari penggunaan RBMwizard untuk pembuatan database RBMware.
- Mendemonstrasikan cara penambahan dan pengubahan Database Manager dengan menggunakan RBMwizard dan Database Setup.
- Mempelajari pembuatan route dengan Route Manager

Pembuatan Database Menggunakan RBMwizard

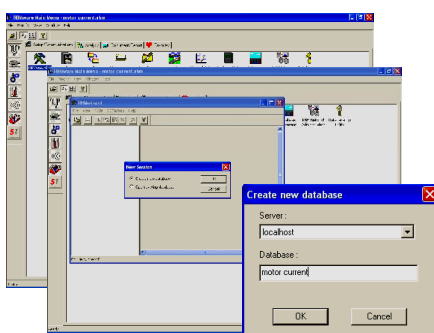
47

- Klik **Next** tampil window default units.
- Pada **General** pilih **English/Metric**
- Pada **Frequency** pilih **Hz**
- Setelah selesai klik **Finis**



Pembuatan Database Menggunakan RBMwizard

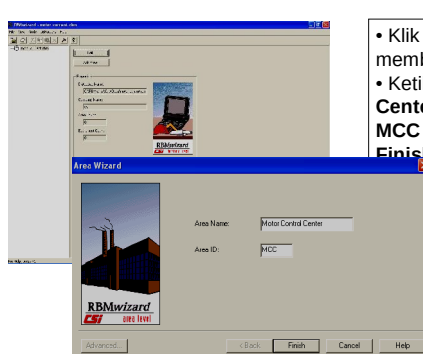
44



- Aktifkan Program RBMware
- Klik ikon **RBMwizard**
- Pilih create new database dan klik **OK**
- Masukkan nama database "**motor current**" dan klik **OK**

Pembuatan Database Menggunakan RBMwizard

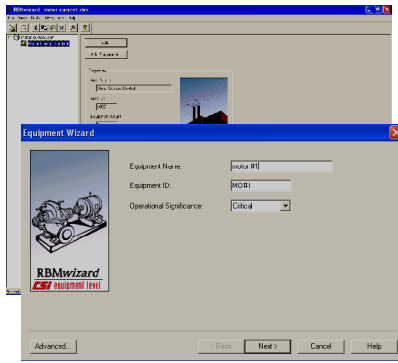
48



- Klik **Add Area** untuk membuat area
- Ketik **Motor Control Center** pada Area Name dan **MCC** pada Area ID, lalu klik **Finish**

Pembuatan Database Menggunakan RBMwizard

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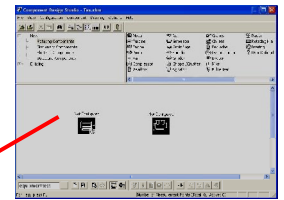
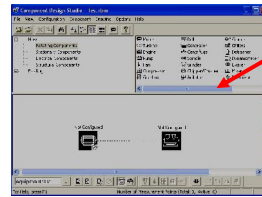


- Pilih **Add Equipment** lalu pada menu Equipment Wizard ketik **Motor #1** pada Equipment Name dan **MO#1** pada Equipment ID, lalu klik **Next**.

Pembuatan Database Menggunakan RBMwizard

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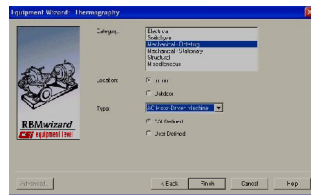
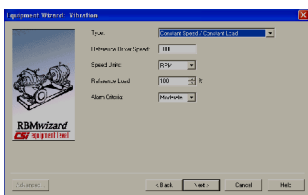
Pada **New (rotating component)** drag motor lalu pompa ke dalam workspace.



Untuk menghubungkan dua komponen tsb klik tombol **coupling** pada bagian bawah, lalu klik drag dari motor ke pompa, sehingga terlihat seperti pada workspace di samping

Pembuatan Database Menggunakan RBMwizard

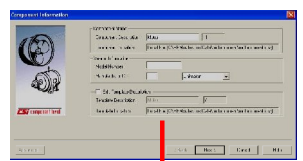
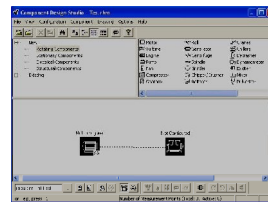
50



- Pilih **Constant Speed / Constant Load** pada Type. Pilih **RPM** pada Speed Units dan **100 %** Load. Select **Smooth** sebagai alarm criteria.
- Thermography screen muncul. Pilih Category **Mechanical - Rotating** dan Tipe **AC motor - Driven Machine**, lalu klik **Finish**.

Pembuatan Database Menggunakan RBMwizard

54

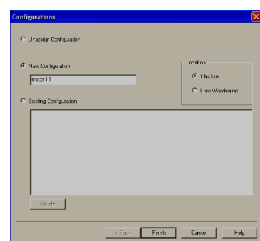
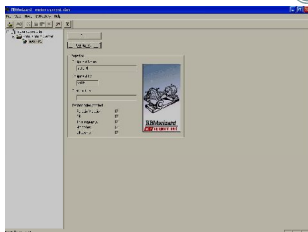


- Untuk membuat konfigurasi motor double klik gambar ikon motor pada workspace

- Pada component information isi data yang anda ketahui lalu klik **Next**

Pembuatan Database Menggunakan RBMwizard

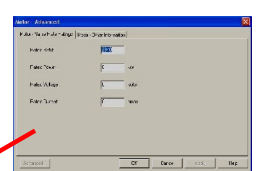
51



- Klik **Configure**. Dan kita akan memasuki Component Design Studio (CDS).
- Pilih **New Configuration** dan **This File** pada **Location**, lalu klik **Finish**.

Pembuatan Database Menggunakan RBMwizard

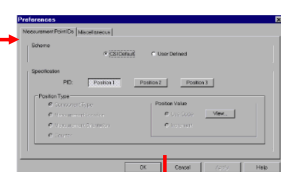
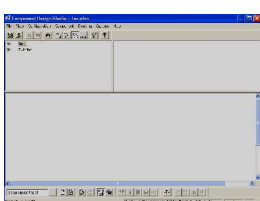
55



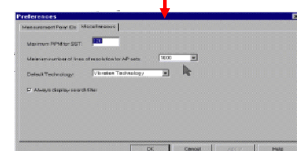
- Pada bagian **Advanced** isi baik data dari name plate maupun pada *other information*
- Setelah selesai klik **OK**

Pembuatan Database Menggunakan RBMwizard

52



- Pada "Miscellaneous" diset default SST minimum RPM, Lines of Resolution, dan Technology. Set Min SST RPM pada **120** dan minimum LOR pada **1600**, lalu klik **OK**.



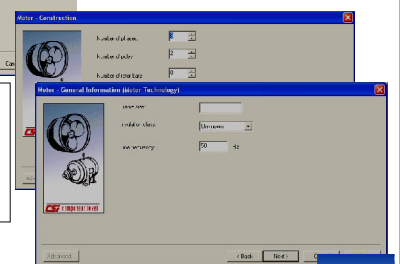
Pembuatan Database Menggunakan RBMwizard

56



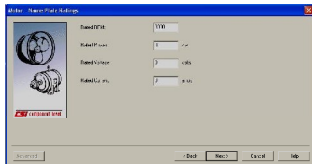
Klik **Next** lalu isi data konstruksi motor yang dipilih, bila selesai klik **Next**

- Isi insulation class atau **Unknown**
- Pada bagian **line frekuensi** diisi dengan **50 Hz (60 Hz)**, lalu klik **Next**



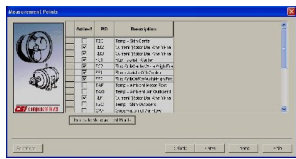
Pembuatan Database Menggunakan RBMwizard

57



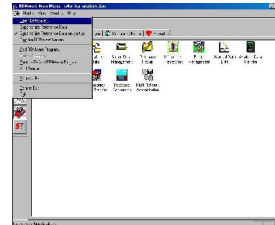
Pilih measurement point yang anda inginkan dengan mengklik pada bagian **Active**?

Bila selesai klik **Finish**.



Membuat Route

61



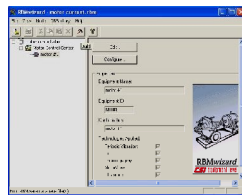
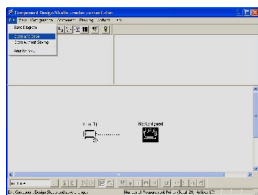
- Aktifkan program RBMware.
- Setelah program RBMware aktif, pilihlah **File>Open Database**.



- Pilihlah database **rotor bar analysis.rbm** kemudian tekan tombol **OK**.
- Kemudian klik ikon **Route Management**.

Pembuatan Database Menggunakan RBMwizard

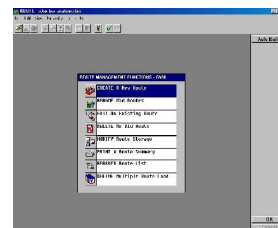
58



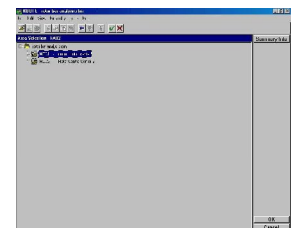
- Setelah dicek maka konfigurasi ini selesai klik **File**, lalu **Close and Save**.
- Pada RBMwizard klik gambar palu (**Built**) sehingga anda telah selesai membuat satu database.

Membuat Route

62



- Setelah masuk ke program **ROUTE** pilihlah **CREATE A New Route**.



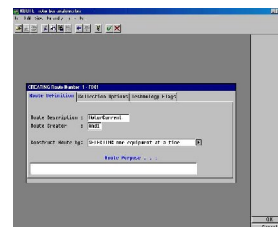
- Pilih area **MCC1**, kemudian tekan tombol **OK**.

Route Management Dan Analyzer Data Transfer RBMware

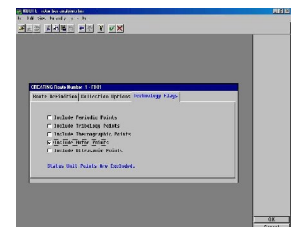
59

Membuat Route

63



- Pada **Route Description** isilah dengan **MotorCurrent**.
- Pada **Route Creator** isilah dengan nama pembuat route.



- Pindahlah ke bagian **Technology Flags**, pilihlah **Include Motor Points**.
- Kemudian tekan tombol **OK**.

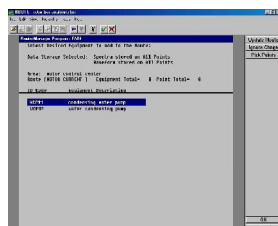
Tujuan

60

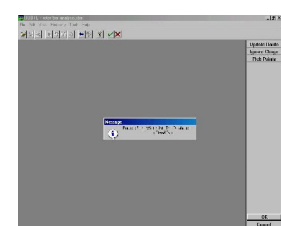
- Mengetahui cara pembuatan route dengan **Route Management**.
- Mengetahui cara mengedit route yang telah dibuat dengan **Route Management**.
- Mengetahui cara penggunaan **Analyzer Data Transfer** untuk transfer data dari **RBMware** ke **CSI 2120** atau sebaliknya.

Membuat Route

64



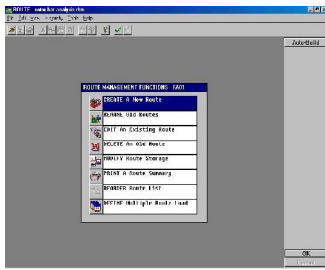
- Setelah itu pilihlah equipment **WCP#1** dengan menekan tombol **OK**.



- Kemudian tekan tombol **Update Route** untuk menulis route yang telah dibuat ke database.

Membuat Route

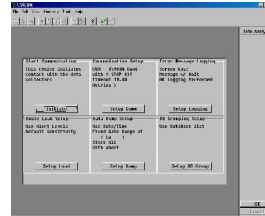
65



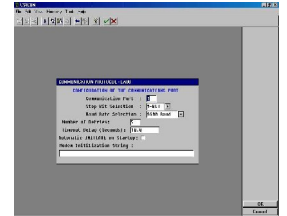
- Setelah muncul tampilan utama program ROUTE, buatlah route untuk area MCC2.
- Pada Route Description isilah dengan MotorFlux.
- Pindahlah ke bagian Technology Flags, pilihlah Include Motor Points.

Menggunakan Analyzer Data Transfer

69



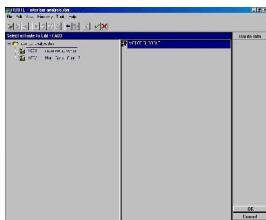
- Untuk masuk ke program Analyzer Data Transfer klik ikonnya pada tampilan utama program RBMware.



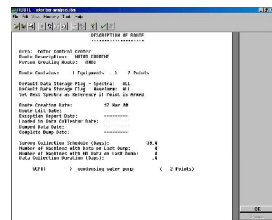
- Pilih Setup Comm untuk mengubah konfigurasi port komunikasi.

Mengedit Route

66



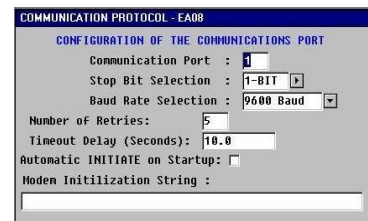
- Pilih Edit An Existing Route.
- Sorot pada route Motor Current, kemudian tekan tombol Route Info.



- Akan muncul informasi mengenai route tersebut.
- Tekan tombol OK untuk kembali ke tampilan awal.

Menggunakan Analyzer Data Transfer

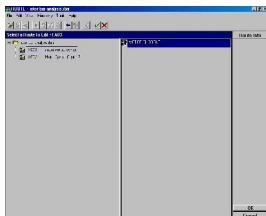
70



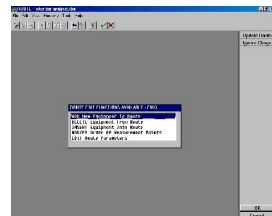
- Sesuaikan nomor Communication Port sesuai dengan port yang akan digunakan pada komputer.
- Samakan pilihan Stop Bit Selection, Baud Rate Selection, Number of Retries dan Timeout Delay dengan setting di CSI 2120.

Mengedit Route

67



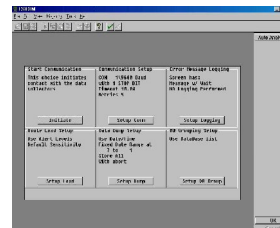
- Pilih kembali route Motor Current dan tekan tombol OK untuk mengedit route tersebut.



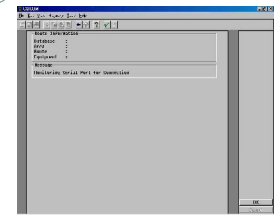
- Untuk menambah equipment pada route Motor Current pilih Add New Equipment To Route kemudian tekan tombol OK.

Menggunakan Analyzer Data Transfer

71



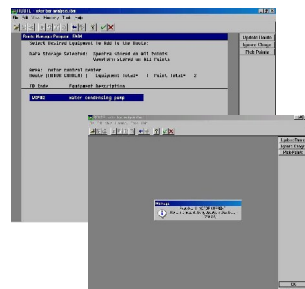
- Untuk kembali ke tampilan utama program CSICOM tekan tombol OK setelah mengonfigurasi port komunikasi.



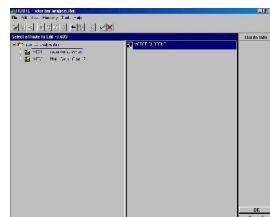
- Kemudian tekan tombol Initiate untuk memulai komunikasi dengan CSI 2120.
- Perintah download/upload data dilakukan melalui CSI 2120.

Mengedit Route

68



- Pilih equipment WCP#2 kemudian tekan tombol OK.



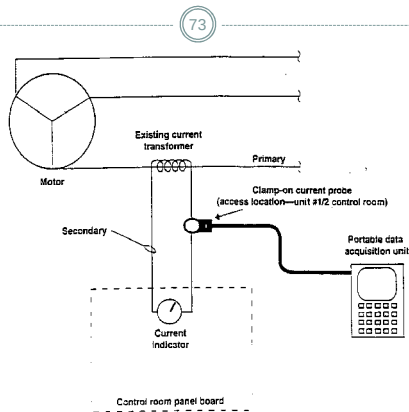
- Hasilnya bisa dicek dengan memilih route Motor Current, kemudian tekan tombol Route Info.

Data Collection

72

MOTOR CURRENT ANALYSIS

Data Collection



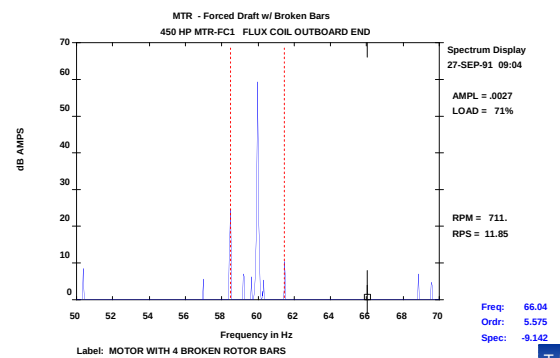
Example

450 HP FORCED DRAFT FAN MOTOR

Application of MCSA, flux coil for Electrical Motor Condition Monitoring (Part 2)

- Data analysis using MotorView program
 - Application of MCSA
 - Rotor Failure
 - Diagnose Rotor Bar Problems.

35 dB down shows broken rotor bars



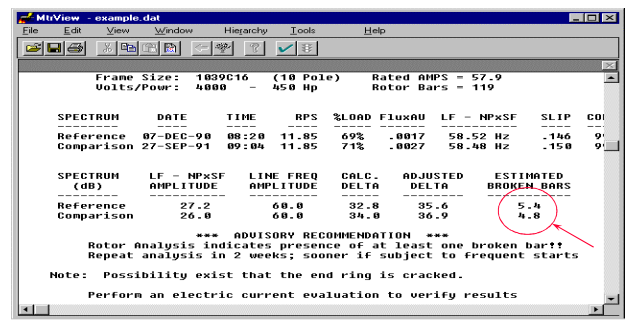
Rotor Bar Analysis

- Can be performed by two methods:

1. Current Measurement
2. Flux Measurement



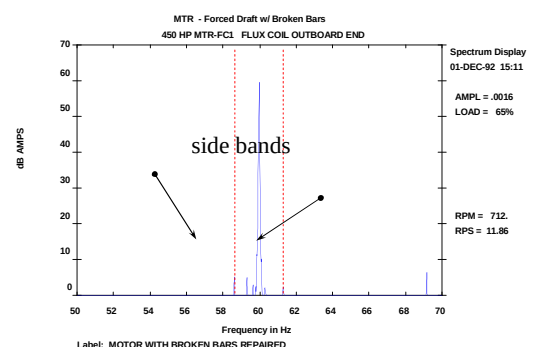
AMS Suite™: Machinery Health™ Manager MotorView II diagnosed data



Parameter tingkat kerusakan rotor by MCSA

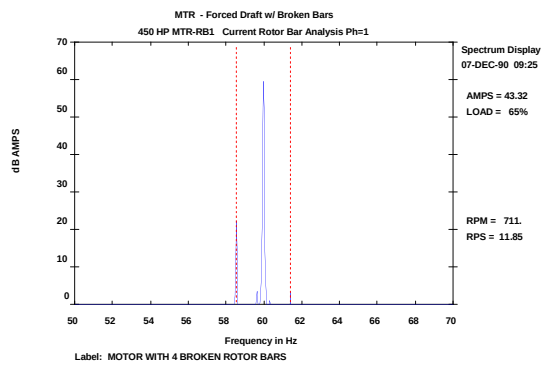
- Selisih dB amplitude antara LF dan LSB1
- 54 dB- Kondisi motor normal.
- 45- 54 dB- Kondisi motor adalah marginal condition
- 40- 45 dB- Setidaknya satu broken rotor bar dapat ditemukan (lakukan trending dan monitor terus)
- <40 dB- Lebih dari satu broken rotor bar. Direkomendasikan untuk mematikan motor bila dalam kondisi ini.
- Reference : Electrical Apparatus Service Association (EASA

Repaired rotor bars on same machine



Bad Rotor Bars on same machine

81



TIARA
Reliability Generation

Flux Monitoring

85

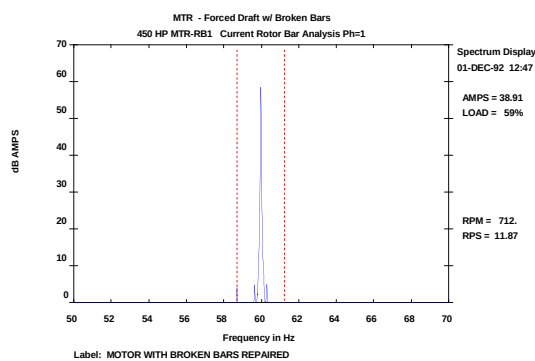
- Use 343 Flux Coil
- Position on the fan shroud of a motor
- Measures magnetic flux signature of AC induction motors
- Detects
 - s - rotor bar faults
 - s - stator faults
 - s - unbalanced voltage
 - s - motor speed



TIARA
Reliability Generation

Good Rotor Bars on same machine

82



TIARA
Reliability Generation

FLUX Notes

86

- Do not need to know load
- Do not need to know # of bars
- Do not need to know # of stator slots
- Need to position coil consistently
- Need MCSA baseline

TIARA
Reliability Generation

Application of MCSA, flux coil for Electrical Motor Condition Monitoring (Part 3)

83

□ Using Flux to Analyze Motor Problems

- Flux Analysis for Rotor Faults
- Flux Analysis for Stator Faults
- Unbalance Voltage Supply
- Shorted Winding Turn
- Shorted rotor lamination
- Phasing problem

TIARA
Reliability Generation

FLUX Benefits

87

- Detects same faults as motor current and more
 - without accessing electric panels
 - without connecting to live power leads
 - without requiring an electrician
 - during vibration route collection

TIARA
Reliability Generation

Diagnosing Rotor Bar Problems.

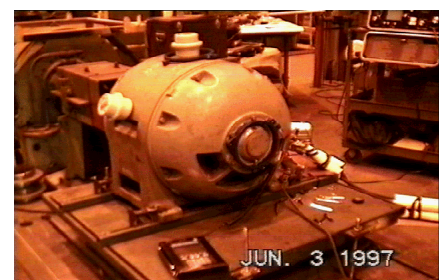
84

- The life of a rotor is directly proportional to the number of starts and stops of a motor.
- To diagnose rotor problems with flux we capture a baseline current reading to normalize flux measurements.
- Then we trend the flux measurement and look for side bands of (slip frequency X #poles) about Line Frequency.

TIARA
Reliability Generation

Using Flux to Analyze Motor Problems

88



Picture Shows Mounting Positions of Flux Coils

TIARA
Reliability Generation

Flux Analysis for Stator Faults

89

- Baseline flux reading mandatory
- Two Measurements required:
 - 1) Low: Trend LF with sidebands of RPS.
 - 2) High: Trend slot pass family Slot Pass Freq. with sidebands of 2xLF
- All readings from the same location
- Watch for change from baseline.

Readings Repeated Under Different Loads (No significant change in values noted)

93

FREQUENCY (Hz)	DESCRIPTION	AMPLITUDE (dB)	REFERENCE	COMPARISON	DIFFERENCE (COMP-REF) (dB)	RMS RATIO COMP/REF
931.9	PSP-12xLF	-24.21	-22.88	-22.88	1.34	1.17
1851.9	PSP-18xLF	-27.14	-28.16	-28.16	-3.01	1.37
1172.0	PSP-8xLF	-25.83	-22.10	-22.10	3.73	1.54
1272.1	PSP-6xLF-RPS	-29.33	-26.23	-26.23	3.10	1.43
1292.0	PSP-6xLF	-16.87	-12.83	-12.83	4.04	1.75
1419.8	PSP-4xLF	-3.40	-1.75	-1.75	1.65	1.83
1532.0	PSP-2xLF	-24.34	-19.92	-19.92	4.42	1.66
1652.2	PSP+8xLF-RPS	1.37	0.91	0.91	-0.46	0.95
1652.1	PSP+8xLF	11.34	12.42	12.42	1.07	1.13
1672.0	PSP+8xLF-RPS	-33.42	-31.40	-31.40	2.02	1.26
1752.2	PSP+2xLF-RPS	-27.50	-26.44	-26.44	1.06	1.13
1772.1	PSP+2xLF	15.25	15.07	15.07	-0.18	0.98
1792.0	PSP+2xLF-RPS	-25.25	-24.29	-24.29	0.95	1.12
1892.2	PSP+4xLF	-18.43	-12.90	-12.90	5.53	1.89
2012.2	PSP+6xLF	-16.93	-16.38	-16.38	0.55	1.07
2112.4	PSP+8xLF-RPS	-38.12	-43.69	-43.69	-5.57	1.18
2132.2	PSP+8xLF	-18.47	-15.31	-15.31	3.16	1.57
2252.3	PSP+18xLF	-18.71	-17.55	-17.55	1.16	1.14

50%
Load

85% Load

Empirical Test of Flux

90

Test Measurement #1

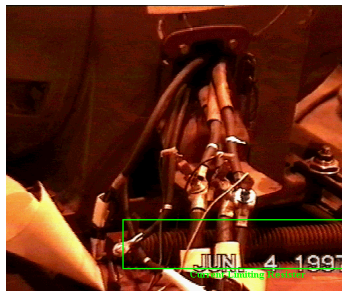
94

INDUCED SMALL SHORT IN STATOR

Empirical Test

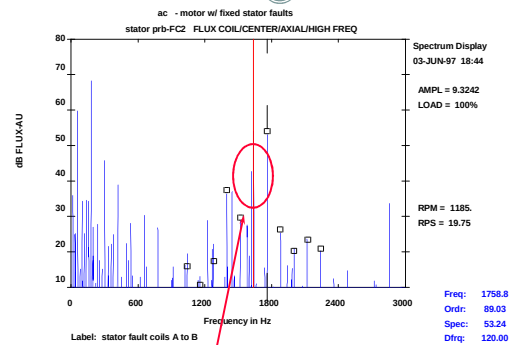
91

- 150 HP Motor
- Wound To Allow Three Levels Of Turn to Turn Shorts



Flux Reading with Small Short

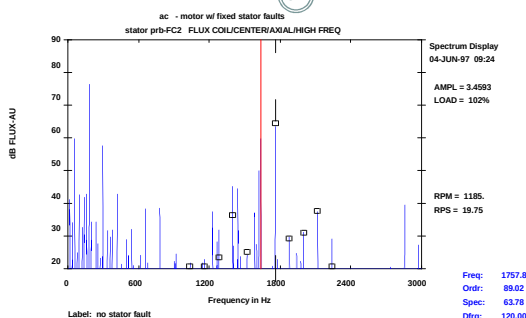
95



Increase in Sideband Values

Baseline Flux Reading

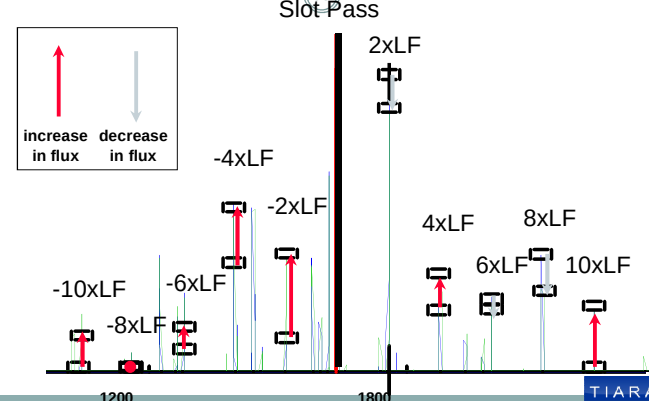
92



Looking for significant change in 120 Hz side band of PSP

Flux Reading - Change from Baseline

96



FREQUENCY (Hz)	AMPLITUDE (dB)	DIFFERENCE (COMP-REF) (dB)	RHS RATIO COMP/REF
VALUE	DESCRIPTION	REFERENCE	COMPARISON
918.6	PSP-12xLF	-24.21	-24.33
1038.6	PSP-18xLF	-27.14	-20.75
1158.7	PSP-8xLF	-25.83	-27.29
1278.7	PSP-6xLF	-16.87	-18.89
1398.8	PSP-4xLF	-3.49	-3.96
1518.8	PSP-2xLF	-24.34	-11.53
1519.2	PSP+8xLF+RPS	1.37	2.45
1638.9	PSP+8xLF	11.34	13.85
1658.7	PSP+8xLF+RPS	-33.42	-29.18
1739.3	PSP+2xLF+RPS	-27.58	2.93
1759.9	PSP+2xLF	15.25	12.90
1879.0	PSP+4xLF	-18.43	3.97
1999.1	PSP+6xLF	-16.93	-1.71
2119.1	PSP+8xLF	-18.47	-15.78
2239.2	PSP+18xLF	-18.71	-18.77

We see 4.3 times increase in flux data

VALUE	DESCRIPTION	REFERENCE	COMPARISON	(COMP-REF) (dB)	COMP/REF	RHS RATIO COMP/REF
918.0	PSP-12xLF	-24.21	-17.63	6.58	2.1	1.17
1039.0	PSP-18xLF	-27.14	-16.02	11.12	3.6	1.71
1159.0	PSP-8xLF	-25.83	-19.73	6.10	2.0	1.54
1279.1	PSP-6xLF	-16.87	-13.28	3.66	1.5	1.43
1399.1	PSP-4xLF	-3.49	-4.95	-1.46	.8	1.75
1418.9	PSP-4xLF+RPS	-37.37	-35.80	1.57	1.2	1.83
1519.2	PSP-2xLF	-24.34	-6.28	18.14	8.0	1.66
1639.3	PSP+8xLF	11.34	19.09	7.64	1.9	1.95
1658.9	PSP+8xLF+RPS	-33.42	-27.89	5.52	1.8	1.13
1739.5	PSP+2xLF+RPS	-27.58	-24.43	3.06	1.4	1.26
1759.3	PSP+2xLF	15.25	13.42	-1.83	.8	1.13
1778.9	PSP+2xLF+RPS	-25.25	-43.33	-18.08	.1	.98
1859.6	PSP+4xLF+RPS	-35.29	-38.33	-3.03	.7	1.12
1879.3	PSP+4xLF	-18.43	-3.30	15.13	5.7	1.89
1979.6	PSP+6xLF+RPS	-25.99	-26.59	-.59	.9	1.07
1999.4	PSP+6xLF	-16.93	-13.85	3.08	1.4	.53
2099.6	PSP+8xLF+RPS	-38.12	-36.58	1.53	1.1	.57
2119.4	PSP+8xLF	-18.47	-12.87	-1.68	.8	1.14
2239.5	PSP+18xLF	-18.71	-22.18	-3.47	.6	

Flux showed 8 times increase over baseline

Test Measurement #2

98

INDUCED MEDIUM SHORT IN STATOR

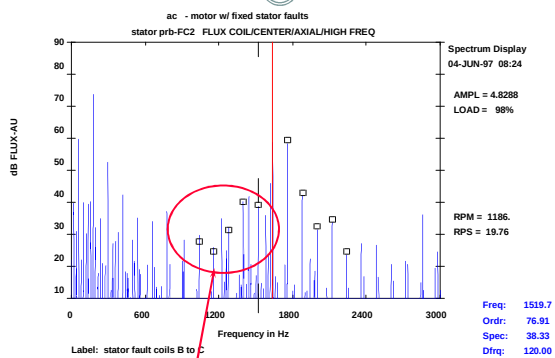
Test Measurement #3

102

INDUCED SEVERE SHORT IN STATOR

Flux Reading with Medium Short

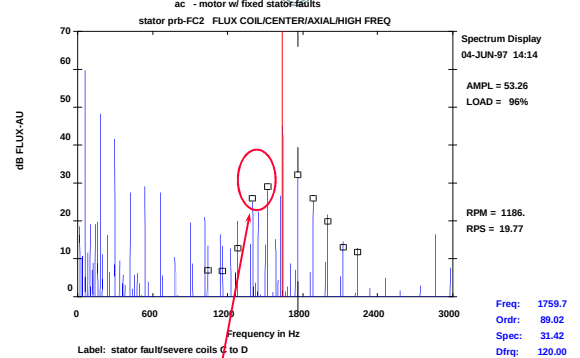
99



Sizeable Increase in Sideband Values

Flux Reading with Severe Short

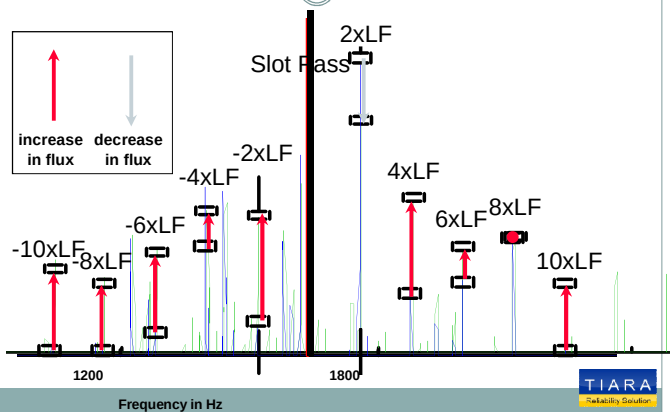
103



Significant Increase in Sideband Value

Flux Reading - Change from Baseline

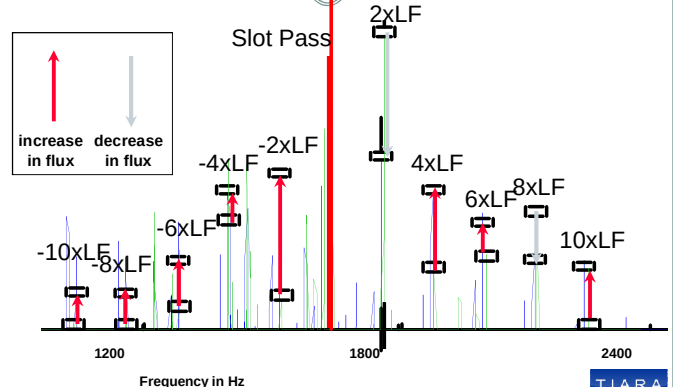
100



Frequency in Hz

Flux Reading - Change from Baseline

104



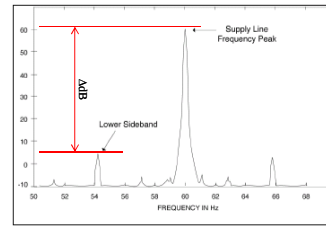
Frequency in Hz

SLOT PASS FAMILY	COMPARISON-REFERENCE + ENERGY - ENERGY	NORMALIZED POSITIVE	COMP-REF NEGATIVE
1839.8	PSP-18xLF	-27.14	-11.71
1159.8	PSP-8xLF	-25.83	-11.61
1268.2	PSP-6xLF-RPS	-29.33	-18.97
1279.8	PSP-6xLF	-16.87	-5.18
1399.8	PSP-4xLF	-3.49	-2.8
1419.6	PSP-4xLF-RPS	-37.37	-31.61
1519.9	PSP-2xLF	-24.34	3.08
1639.9	PSP-8xLF	11.34	28.26
1659.6	PSP-8xLF-RPS	-33.42	-23.43
1748.2	PSP-2xLF-RPS	-27.58	-18.95
1759.9	PSP-2xLF	15.25	9.25
1868.3	PSP-4xLF-RPS	-35.29	-18.38
1879.9	PSP-4xLF	-19.43	-88
1988.3	PSP-4xLF-RPS	-25.99	-15.62
1999.9	PSP-6xLF	-16.93	-3.19
2019.6	PSP-6xLF-RPS	-44.15	-32.61
2188.3	PSP-8xLF-RPS	-38.12	-19.04
2128.8	PSP-8xLF	-18.47	-10.48
2248.8	PSP-18xLF	-18.71	-11.71

23 X increase shown here

Parameter Kerusakan Rotor Bar

109

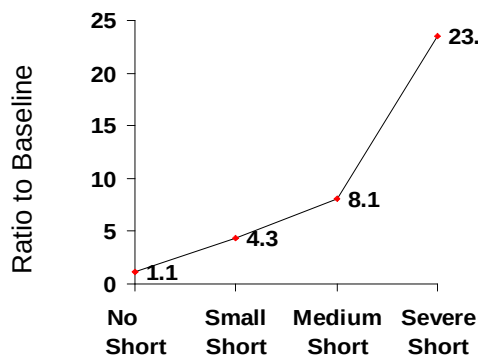


Jika Δ dB :

- >54 dB = Good Motor
- 45 – 54 = marginal condition
- 40 – 45 = setidaknya satu broken rotor
- < 40 = - Lebih dari satu broken rotor bar

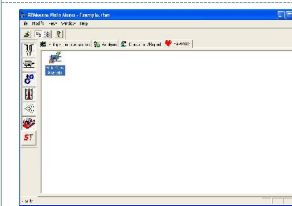
Trend of 2xLF Sideband of Slot Pass

106

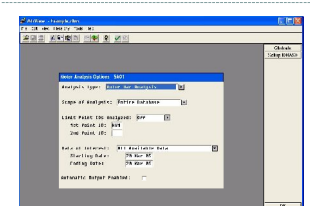


Menggunakan Rotor Bar Analysis

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- Pada RBMware pilih Motorview Analysis
- Pada MotorView Analysis pilih **Rotor Bar Analysis** pada Analysis Type



- Pilih **Entire Database** pada Scope of Analysis
- Pada limit Point Ids Analyzed pilih Off
- Pilih **All Available Data** pada Data of Interest, lalu klik menu **Global** pada Function Bar

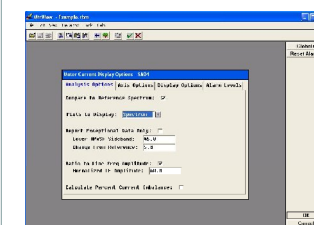
Building report

107

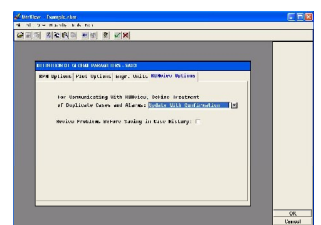
DENGAN MOTORVIEW SOFTWARE

Menggunakan Rotor Bar Analysis

111



- Pada Analysis Option pilih Point to Display ; **Spectrum**
- Klik Ratio to Line freq amplitude jika anda ingin amplitude LF dinormalize menjadi nilai tertentu (misalnya **60 dB**)

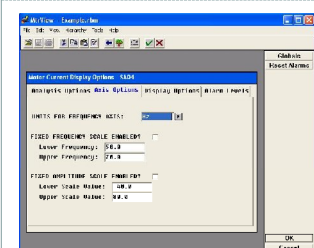


- Sebelum mengisi semuanya set dulu Menu global pada function bar di sebelah kanan.
- Pada RBMview Option pilih **Update With Confirmation** Klik OK

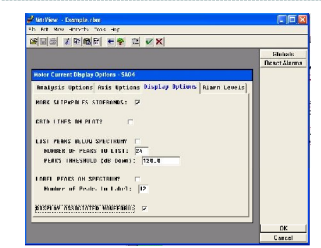
Rotor Bar Analysis

Menggunakan Rotor Bar Analysis

112

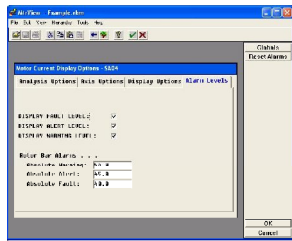


- Pada Display Option pilih Yes pada **Mark SlipxPoles Sidebands**
- Klik yes untuk Display Associated Waveform

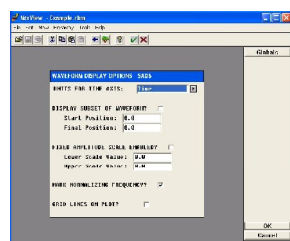


- Pada Axis Option Pilih Unit for Frequency axis; **Hz**
- Jika tidak diinginkan skala yang fix maka biarkan bagian lain kosong

Menggunakan Rotor Bar Analysis

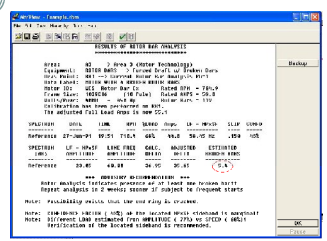
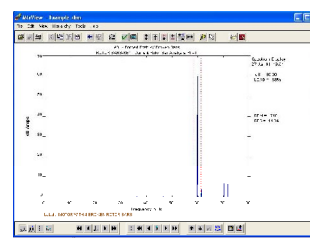


- Pilih alarm option
- Masukan angka-angka alarm standard.



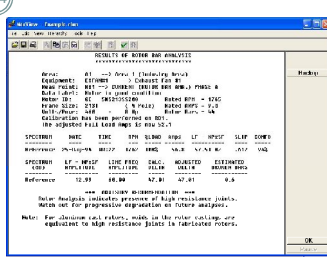
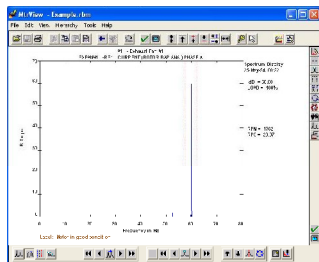
- Pilih **Time** pada **Unit for Time Axis**
- Pilih **yes** pada **Mark Normalizing Frequency**

Menggunakan Rotor Bar Analysis



Pada equipment yang lain terlihat bahwa ada sideband di sekitar line frekuensi
Dengan tingkat kepercayaan 43 % kemungkinan ada bar yang patah

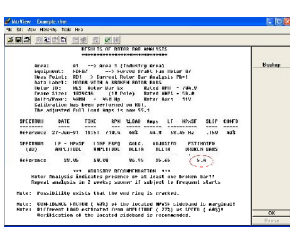
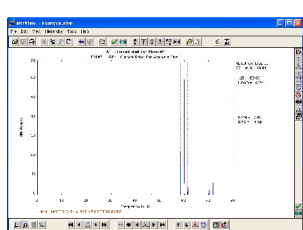
Menggunakan Rotor Bar Analysis



- Dari data terlihat Confidence = 94 % dan estimated Broken Bar = 0,6.
- Disimpulkan bahwa motor dalam kondisi baik

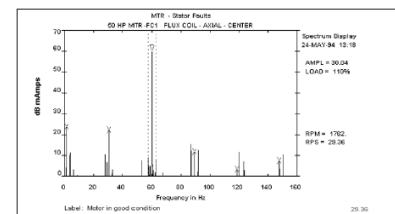
Flux Coil Analysis

Menggunakan Rotor Bar Analysis



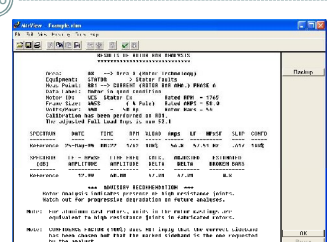
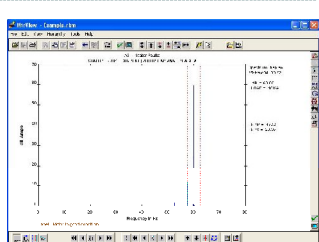
- Pilih Equipment yang lain
- Pada summary terlihat bahwa estimated Broken Bar adalah 5,4 dengan kepercayaan 43 %.
- kemungkinan ada bar yang patah

Sinyal – Sinyal Hasil Pengukuran Flux Coil



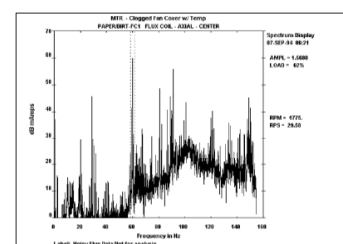
- Sinyal kondisi normal.
- Garis putus-putus mengindikasikan NpxSF sideband di sekitar line frekuensi
- Peak yang disilang adalah runing speed di sekitar line frekuensi.

Menggunakan Rotor Bar Analysis



Hasil pengukuran di area yang lain (pada stator) terlihat bahwa tidak ada sideband di sekitar Line Frekuensi

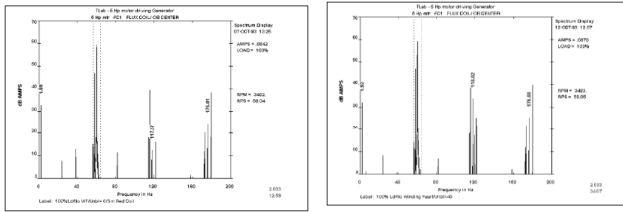
Sinyal – Sinyal Hasil Pengukuran Flux Coil



Sinyal yang diakibatkan oleh ketidakbenaran penempatan alat ukur sehingga terpengaruh oleh vibrasi

Unbalance Voltage Supply

121

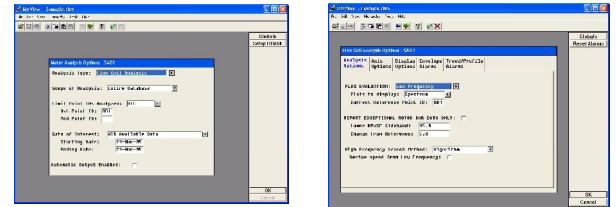


Spektrum flux 50 HP motor dengan 0% unbalance

Spektrum flux 50 HP motor dengan 3% unbalance

Memilih Motor Analisis Option

125



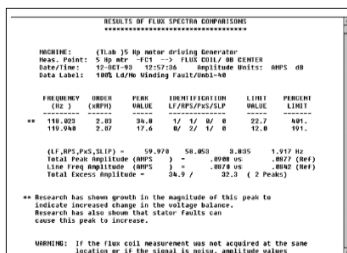
Pilih flux coil analysis pada Analysis Type

Pada analysis Option

- Flux Evaluation dipilih **Low Frequency**
- lalu pilih **spectrum** pada plot to display
- Measurement Point adalah **RB1**

Unbalance Voltage Supply

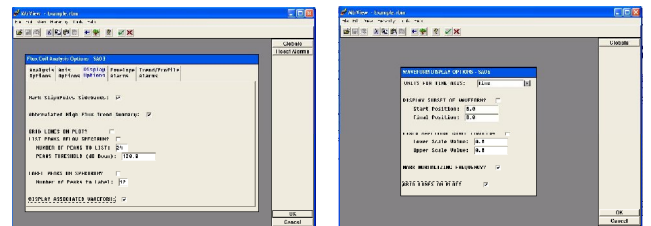
122



Hasil perbandingan spektrum flux

Memilih Motor Analisis Option

126

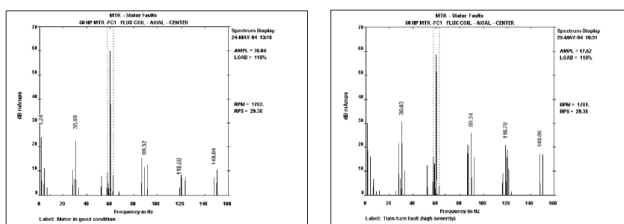


Dalam Display Option klik yes pada Mark SlipxPoles Sideband, Abbreviated high flux summary serta wave form, lalu klik OK

Pada Waveform pilih Time untuk time axis, serta yes untuk Mark normalizing freq dan grid lines on plot, lalu klik OK

Shorted Winding Turn

123

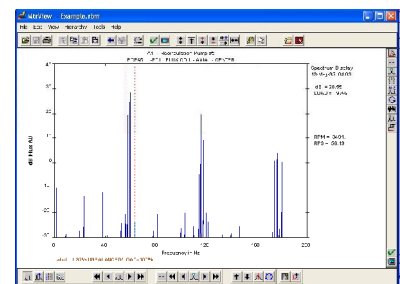


Motor 50 HP dengan stator dalam kondisi baik

Motor 50 HP dengan short pada stator winding

Memilih Motor Analisis Option

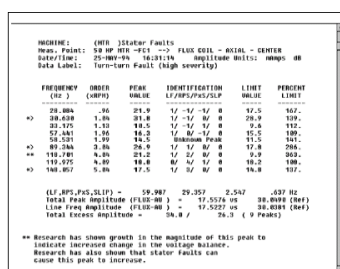
127



Muncul plot yang siap dianalisis

Shorted Winding Turn

124



Perbandingan antara 2 spektrum yang dihasilkan

Terima Kasih
End

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